

Solution to MP363 Problem Set 3

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P. 1

We have two proofs of this - one using operators, and one using the QM continuity eq'n $\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$ (with $\rho = |\psi|^2$ and $\vec{J} = \frac{i\hbar}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi)$). Now let's do the former.

$$|\psi|^2 = \psi^* \psi, \text{ so } \frac{d}{dt} \int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} |\psi(x)|^2 dx$$

$$= \int_{-\infty}^{\infty} \frac{\partial}{\partial t} (\psi^* \psi) dx = \int_{-\infty}^{\infty} \left(\frac{\partial \psi^*}{\partial t} \psi + \psi^* \frac{\partial \psi}{\partial t} \right) dx$$

But the SE gives an expression for $\frac{\partial \psi}{\partial t}$:

$$\frac{\partial \psi}{\partial t} = \frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi \right) = \frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V\psi$$

But because t is a real variable, we can get $\frac{\partial \psi^*}{\partial t}$ by complex-

conjugating this:

$$\frac{\partial \psi^*}{\partial t} = \left(\frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V\psi \right)^* = -\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V\psi^*$$

(because $\hbar, m, x$ and V are all real). We can then use these

two the derivative in the above:

$$\frac{\partial \psi^*}{\partial t} \psi + \psi^* \frac{\partial \psi}{\partial t} = \left(-\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V\psi^* \right) \psi + \psi^* \left(\frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V\psi \right)$$

$$= -\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} \psi + \frac{i}{\hbar} V|\psi|^2 + \frac{i\hbar}{2m} \psi^* \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V|\psi|^2$$

$$= \frac{i\hbar}{2m} \left(\psi^* \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} \psi \right)$$

But in problem 4a of PS 2, we showed that, for any two functions f and g ,

$$f \frac{\partial g}{\partial x} - \frac{\partial f}{\partial x} g = \frac{\partial}{\partial x} \left[f \frac{\partial g}{\partial x} - \frac{\partial f}{\partial x} g \right].$$

So if $f = \psi^*$ and $g = \psi$, we get

$$\frac{\partial \psi^*}{\partial t} \psi + \psi^* \frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m} \frac{\partial}{\partial x} \left[\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right].$$

Now, $\frac{d}{dt} \int_{-\infty}^{\infty} |\psi|^2 dx$ is the integral of this, so

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\psi|^2 dx = \frac{i\hbar}{2m} \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \left[\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right] dx.$$

But integration is the inverse operation to differentiation! The fundamental

theorem of calculus says that

$$\int_a^b \frac{dF(x)}{dx} dx = F(b) - F(a)$$

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\psi|^2 dx = \frac{i\hbar}{2m} \left[\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right]_{-\infty}^{\infty}$$

But for physical states, the condition that their inner product exists

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requires all states to vanish at both $x = +\infty$ and $x = -\infty$. Since ψ is assumed to be such a state, ψ and ψ^* both are zero at both the upper and lower limits of this definite integral. Thus, we conclude that

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\psi(t,x)|^2 dx = 0 \Rightarrow \int_{-\infty}^{\infty} |\psi(t,x)|^2 dx \text{ is time-independent}$$

and so if we set it to 1 at any time (e.g. $t=0$), it's equal to 1 for all times. This allows us to normalize all states to have unit length.

P. 2

In each of the three cases, we state what the inner product is for the situation at hand and how our states go into it.

We just put the same state into both slots together $\langle \psi | \psi \rangle$, which is then set equal to 1 to get $\|\psi\| = 1$. We then determine what value each constant must have in order to make this true.

(a) α_1 is real and positive, so $\psi^* = \alpha_1 \sqrt{x} e^{-x}$, and so

$$\langle \psi | \psi \rangle = \int_0^{\infty} (\alpha_1 \sqrt{x} e^{-x}) (\alpha_1 \sqrt{x} e^{-x}) dx$$

$$= \alpha_1^2 \int_0^{\infty} x e^{-2x} dx$$

$$= \alpha_1^2 \left[-\frac{1}{2} x e^{-2x} + \frac{1}{2} \int_0^{\infty} e^{-2x} dx \right]$$

$$= \alpha_1^2 \left\{ 0 + \frac{1}{2} \left[-\frac{1}{2} e^{-2x} \right]_0^{\infty} \right\} = \frac{\alpha_1^2}{4}$$

so if $\langle \psi | \psi \rangle = 1$, then $\boxed{\alpha_1 = 2}$. (Since we let $\alpha_1 > 0$, we don't have to include -2 as a possibility.)

(b) $\langle u | u \rangle = u_1^* u_1 + u_2^* u_2 = |u_1|^2 + |u_2|^2$, so since

$$u_1 = 2i\alpha_2 \text{ and } u_2 = -i\alpha_2,$$

$$\langle u | u \rangle = |2i\alpha_2|^2 + |-i\alpha_2|^2 = 4\alpha_2^2 + \alpha_2^2 = 5\alpha_2^2$$

so if $\langle u | u \rangle = 1$, then $\alpha_2^2 = 1/5$ and thus $\boxed{\alpha_2 = 1/\sqrt{5}}$.

(Also, $-1/\sqrt{5}$ is ~~eliminated~~ because $\alpha_2 > 0$.)

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(c) Here we use a basic fact that's worth remembering: if a state $|\psi\rangle$ is expanded in an orthonormal basis $|\phi_n\rangle$, then $|\psi\rangle = \sum_n c_n |\phi_n\rangle$ implies $\sum_n |c_n|^2 = 1$ if $\langle\psi|\psi\rangle = 1$. Here, we're told that $|\phi_n\rangle$ is such a basis (that's what $\langle\phi_m|\phi_n\rangle = \delta_{mn}$ means), and thus, if we can determine the expansion coefficients c_n and set $\sum_n |c_n|^2$ to 1, we'll find α_3 .

Well, $c_n = \frac{(-1)^n \alpha_3}{3^n}$ is the coefficient of $|\phi_n\rangle$, so we've set the coefficient, n runs from 0 to ∞ , so our normalization condition

$$\sum_{n=0}^{\infty} |c_n|^2 = \sum_{n=0}^{\infty} \left| \frac{(-1)^n \alpha_3}{3^n} \right|^2 = \sum_{n=0}^{\infty} \frac{\alpha_3^2}{9^n} = \alpha_3^2 \sum_{n=0}^{\infty} \frac{1}{9^n}.$$

This is a geometric series of the form $a + ar + ar^2 + ar^3 + \dots$

$$= \sum_{n=0}^{\infty} ar^n \text{ with } a = \alpha_3^2 \text{ and } r = 1/9. \text{ Since } |r| < 1,$$

it converges to $\frac{a}{1-r}$, or $\frac{\alpha_3^2}{1-1/9} = \frac{9\alpha_3^2}{8}$. Thus, for this to be 1, $\boxed{\alpha_3 = 2\sqrt{2}/3}$ must be the (real and positive) constant.

P.3

Now we take the same three states and inner products and use them to compute the expectation values of three observables. Recall that $\langle A \rangle = \langle\psi|A|\psi\rangle = \langle\psi|(A|\psi\rangle)$. So we act A on $|\psi\rangle$, then take the inner product of the result with $\langle\psi|$. Here's how it goes.

(a) P acts on functions as $P\psi(x) = -i\hbar \frac{\partial \psi(x)}{\partial x}$. For the function we have here

$$P\psi(x) = -i\hbar \frac{\partial}{\partial x} [\alpha_1 \sqrt{x} e^{-x}] = -i\hbar \alpha_1 \left[\frac{1}{2\sqrt{x}} e^{-x} - \sqrt{x} e^{-x} \right].$$

We now take the inner product of this with ψ , i.e.

$$\begin{aligned} \langle P \rangle &= \int_{-\infty}^{\infty} \psi^* (P\psi) dx \\ &= \int_0^{\infty} \alpha_1 \sqrt{x} e^{-x} \left(-\frac{i\hbar \alpha_1}{2\sqrt{x}} e^{-x} + i\hbar \alpha_1 \sqrt{x} e^{-x} \right) dx \\ &= -\frac{i\hbar \alpha_1^2}{2} \int_0^{\infty} e^{-2x} dx + i\hbar \alpha_1^2 \int_0^{\infty} x e^{-2x} dx. \end{aligned}$$

we already found that $\int_0^\infty x e^{-2x} dx = 1/4$, and $\int_0^\infty e^{-2x} dx = -\frac{1}{2} e^{-2x} \Big|_0^\infty = 1/2$, so we get

$$\langle p \rangle = -\frac{i\hbar\alpha_1^2}{2} \left(\frac{1}{2} \right) + i\hbar\alpha_1^2 \left(\frac{1}{4} \right) = \boxed{0}$$

so this state has an average momentum of zero.

(b) U is a column vector, so acting it with the matrix L_z gives

$$L_z U = \frac{\hbar\alpha_2}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2i \\ -i \end{pmatrix} = \frac{\hbar\alpha_2}{2} \begin{pmatrix} 2i \\ i \end{pmatrix}$$

so $(L_z U)_1 = i\hbar\alpha_2$, $(L_z U)_2 = \frac{i\hbar\alpha_2}{2}$. Now, its inner product with U is

$$\begin{aligned} \langle L_z \rangle &= U_1^* (L_z U)_1 + U_2^* (L_z U)_2 = (2i\alpha_2)^* (i\hbar\alpha_2) + (-i\alpha_2)^* \left(\frac{i\hbar\alpha_2}{2} \right) \\ &= \frac{3\hbar\alpha_2^2}{2} = \frac{3\hbar^2}{10} \end{aligned}$$

(c) Again, we now use a fact we derived in lecture; if $|\phi_n\rangle$ are not only an orthonormal basis, but also the eigenvectors of an observable A with $A|\phi_n\rangle = a_n|\phi_n\rangle$, then if a state $|\psi\rangle$ is expanded as $\sum_n c_n |\phi_n\rangle$, then $\langle A \rangle = \sum_n |c_n|^2 a_n$ is the expectation value of A .

That's what we have here; P2(c) told us they were orthonormal, and this problem tells us they're energy eigenstates with $E_n = (-2)^n E_0$. We already know that $c_n = \frac{(-1)^n \alpha_3}{3^n}$, so this tells us

$$\begin{aligned} \langle H \rangle &= \sum_{n=0}^{\infty} |c_n|^2 E_n = \sum_{n=0}^{\infty} \left| \frac{(-1)^n \alpha_3}{3^n} \right|^2 (-2)^n E_0 \\ &= \sum_{n=0}^{\infty} \left(-\frac{2}{9} \right)^n \alpha_3^2 E_0 \\ &= \frac{\alpha_3^2 E_0}{1 - (-2/9)} = \boxed{\frac{9}{11} \alpha_3^2 E_0 = \frac{8}{11} E_0} \end{aligned}$$

P.4

(a) This is easy to see:

$$[AB, C] = (AB)C - C(AB) = ABC - CAB.$$

Now add zero to this in the form $ACB - ACB$:

$$[AB, C] = (ABC - CAB) + (ACB - ACB)$$

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$$= (ABC - ACB) - (CAB - ACB)$$

$$= A(BC - CB) - (CA - AC)B$$

$$= A[B, C] - [C, A]B$$

$$= \boxed{A[B, C] + [A, C]B}$$

because $[A, C] = -[C, A]$.

(b) We'll now use this in the next part: first, we know that $[X, P] = i\hbar 1$.

Now we propose that $[X^n, P] = i\hbar n X^{n-1}$. Let's do this by induction:

(i) Is the proposition true for $n=1$? Putting 1 in for n gives

$$[X^1, P] = i\hbar 1 \cdot X^0. \quad X^1 = X \text{ and } X^0 = 1 \text{ identity operator,}$$

so $[X, P] = i\hbar 1$, in agreement with the fundamental

relation. Thus, it's true for $n=1$.

(ii) Assume it's true for a particular n . Is it then true for

$n+1$? Well, since $X^{n+1} = X^n \cdot X$, we have

$$[X^{n+1}, P] = [X^n \cdot X, P]$$

$$= X^n [X, P] + [X^n, P] X$$

where we have used our result from (a) with $A = X^n$, $B = X$ and $C = P$.

We know that $[X, P] = i\hbar 1$, and we also have as given that

$$[X^n, P] = i\hbar n X^{n-1}$$

$$\begin{aligned} [X^{n+1}, P] &= X^n (i\hbar 1) + (i\hbar n X^{n-1}) X \\ &= i\hbar X^n + i\hbar n X^n = i\hbar (n+1) X^n \end{aligned}$$

which is what our proposal gives. Thus, (true for n) implies

(true for $n+1$)

Thus, we have proved that $[X^n, P] = i\hbar n X^{n-1}$ holds for $n=1, 2, 3, \dots$

The exact same for $[X, P^n] = i\hbar n P^{n-1}$

(i) $n=1$ on both sides gives $[X, P^1] = i\hbar 1 \cdot P^0$, or

$[X, P] = i\hbar 1$, which is known to be true.

(ii) Assume $[X, P^n] = i\hbar n P^{n-1}$. Since $P^{n+1} = P^n \cdot P$, we have

$$[X, P^{n+1}] = -[P^n P, X]$$

$$= -(P^n [P, X] + [P^n, X] P)$$

$$= P^n [X, P] + [X, P^n] P$$

$$= P^n (i\hbar 1) + (i\hbar n P^{n-1}) P$$

$$\textcircled{6} \\ = i\hbar P^n + i\hbar n P^n = i(n+1)\hbar P^n$$

which agrees with our proposal.

Thus, $[X, P^n] = i\hbar n P^{n-1}$ was for all $n=1, 2, 3, \dots$